

1. Determine the domain of definition of f then simplify, its expression by using trigonometric relations in each of the following cases :

i) $f(x) = \sin(\arccos(x))$, **ii)** $f(x) = \cos(\arcsin(x))$,
iii) $f(x) = \cos(\arctg(x))$, **iv)** $f(x) = \sin(\arctg(x))$

2. Calculate : $\arcsin(\sin a)$, $\arccos(\cos a)$, $\arctan(\tan a)$, $\arccos(\sin a)$, for $a \in \left\{\frac{61\pi}{5}, \frac{76\pi}{5}, \frac{83\pi}{5}\right\}$.

3. (1.) Show that : $0 < \arccos(3/4) < \frac{\pi}{4}$. (2.) Solve : $\arccos x = 2 \arccos(3/4)$.

4. Let be the function $\cos : [2\pi, 3\pi] \rightarrow [-1, 1]$. Is it bijective? If affirmative, calculate its inverse.

5. Determine the domain of definition of f then, by calculating the derivative find a simple expression of f in each of the following cases :

i) $f(x) = \arctg(x + \sqrt{1+x^2})$, **ii)** $f(x) = \arctg(x) + \arctg\left(\frac{1}{x}\right)$,
iii) $f(x) = \arcsin(x) + \arccos(x)$,

6. Solve in $\mathbb{R}$ the equation $\arcsin(2x) - \arcsin(\sqrt{3}x) = \arcsin x$, and $5chx - 4shx = 3$.

7. Calculate

$$\arcsin\left(\frac{4}{5}\right) + \arcsin\left(\frac{3}{5}\right), \arctg\left(\frac{1}{2}\right) + \arctg\left(\frac{1}{3}\right), \arctg(2) + \arctg(3).$$

8. Determine the domain of definition of f then simplify its expression in each of the following cases :

i) $f(x) = ch(\arg sh(x))$, **ii)** $f(x) = sh(\arg ch(x))$ **iii)** $f(x) = ch(\arg thx)$,
iv) $f(x) = Argsh(2x\sqrt{1+x^2})$, **v)** $f(x) = \arg ch(2x^2 - 1)$, **vi)** $f(x) = \arg sh(3x + 4x^2)$.

9. Find the domain of definition D_f of f and the domain E_f on which the function f is differentiable then calculate the derivative of f for all $x \in E_f$:

$$f(x) = \arcsin\left(\frac{1-x^2}{1+x^2}\right), f(x) = e^x \arcsin x.$$

$$f(x) = \ln^2 x - \ln(\ln x), f(x) = \frac{1+x}{1+\sqrt{x}} \exp\{\arctg x\}.$$

10. Calculate the derivatives of the following functions

$$f(x) = \frac{\sin x + \cos x}{\sin x - \cos x}, f(x) = \frac{(1+x^2) \arctg x - x}{2},$$

$$f(x) = \frac{\arg sh x}{x}, f(x) = \frac{\operatorname{arccotg} x}{\sqrt{1-x^2}}, f(x) = \sin(\cos^2(\tan^3 x)),$$

It can help you.

Practise Mathematics with Benbachir, elementary fuctions playlist. Youtube Channel.